

HAFIZ M. HASSAN

AI Engineer · Data Scientist · Full-Stack Developer

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#) | [Kaggle](#) | [QuickSite](#)

PROFESSIONAL SUMMARY

Data Science student at FAST NUCES — Pakistan's premier CS institution — building at the intersection of AI, machine learning, and production-grade web engineering. Researcher on NLP2Shell, a fine-tuned on-device language model that translates natural language into shell commands. Co-founder of QuickSite, a web agency helping founders launch fast. Passionate about writing clean, efficient code and turning data into impactful digital experiences.

EDUCATION

BS Data Science

FAST NUCES, Lahore — Pakistan's top-ranked CS institution

2024 – Present

Intermediate in Computer Science (ICS)

Government College University (GCU), Lahore

2022 – 2024

Matriculation

UGI, Lahore

2020 – 2022

EXPERIENCE

Co-Founder & Lead Developer · [QuickSite](#) · Lahore, Pakistan (Remote)

2024 – Present

- Co-founded a web agency focused on helping founders launch production-ready websites rapidly with zero monthly lock-ins.
- Architected the full deployment pipeline on Vercel with 100% automation — from prototype to live in one click.
- Designed lightweight, SEO-optimised sites engineered for performance; delivered personalised client experiences across engagements.
- Manage end-to-end client relationships: scoping, design, delivery, and ongoing support.

Junior Game Development Trainer · [FSDS \(Freelancing & Skill Development Society\), GCUL](#) ·

Lahore, Pakistan

Aug 2022 – Jul 2023

- Delivered structured game development and MERN stack training programmes to cohorts of university students.
- Mentored learners from zero to project-ready, building confidence in web development and software engineering fundamentals.
- Coordinated with department heads to align training goals with institutional strategic objectives.

- Strengthened communication and professional presentation skills through repeated live demonstrations.

CURRENT RESEARCH

NLP2Shell — Natural Language → Shell Command Execution

Fine-tuning a compact language model (Qwen2.5-0.5B) with LoRA on 26,000 NL→Bash pairs to let any user describe file operations in plain English and receive a safe, executable shell command — entirely on-device, no cloud required.

- Model: Qwen2.5-0.5B-Instruct · LoRA (r=16) · PEFT · 4-bit NF4 quantization (BitsAndBytes)
- Training: TRL SFTTrainer · Kaggle T4 GPU · Dataset: NL2Bash (26k pairs)
- Speech input via OpenAI Whisper (tiny) — CPU inference, ~1–3s latency
- Multi-layer safety system: hard blocklist · path validation · mandatory confirm [y/n] with audit log

SELECTED PROJECTS

Full-Stack Web

TravelBuddy+ — Travel Companion Platform [🔗 GitHub](#)

Full-stack social travel app featuring a buddy-matching algorithm, trip planner with budget estimator, and a Vloggers Hub for community content.

Stack: React · Node.js · Express · SQL Server

CADS-Bridge — Real-Time CA-DS Collaboration Tool [🔗 GitHub](#)

Enterprise web app bridging Chartered Accountants and Data Science teams with role-based dashboards, an annotation engine, knowledge hub, KPI tracking, and automated conflict detection.

Stack: React · Node.js · Express · SQL Server

Machine Learning & Data Science

Sentiment Predictor [🔗 GitHub](#)

End-to-end NLP sentiment analysis pipeline using TF-IDF vectorisation and ensemble ML models, trained and evaluated on large textual review datasets.

Stack: Python · Scikit-Learn · Pandas · NLP

RecommendationX — Movie Recommender [🔗 GitHub](#)

Browser-based personalised movie recommender using collaborative filtering with a clean JavaScript frontend.

Stack: JavaScript · Node.js · Collaborative Filtering

Systems & Low-Level Engineering

Library Management System [🔗 GitHub](#)

High-performance LMS in C++ implementing AVL trees and hash maps to achieve O(log n) search and insert operations.

Stack: C++ · DSA · Algorithms

Brick Breaker & Space Shooter [🔗 GitHub](#)

Two complete arcade games written in 8086 Assembly using raw video memory manipulation and custom interrupt handling — zero abstractions.

Stack: Assembly x86 · Game Dev

TECHNICAL SKILLS

Languages

Python · JavaScript · C++ · C# · C · Assembly
x86

AI / ML

TensorFlow · Keras · Scikit-Learn · Pandas ·
NumPy · NLP

Research & Tools

PEFT / LoRA · BitsAndBytes · TRL · OpenAI
Whisper · Jupyter

Web

React · Node.js · Express · HTML5 / CSS3 ·
REST APIs

Databases & Infra

SQL Server · Git · GitHub · Vercel

Other

Unity (Game Dev) · VS Code · MS Office · Figma

ONLINE PRESENCE

GitHub

github.com/HassanNawaz14

Kaggle

[hassannawaz1423](https://kaggle.com/hassannawaz1423)

LinkedIn

[hafiz-m-hassan](https://www.linkedin.com/in/hafiz-m-hassan)

LeetCode

[hassan_nawaz](https://leetcode.com/hassan_nawaz)

Fiverr

[hassandev](https://www.fiverr.com/hassandev)

Portfolio

[hassan-nawaz-
portfolio00.vercel.app](https://hassan-nawaz-portfolio00.vercel.app)